RÉSUMÉ

Alec *Layton.*

Software Engineer · Backend Systems · Automation · AI Agent Tooling

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Software engineer with 3+ years building and owning mission-critical software across the full development lifecycle, with a focus on automation, data pipelines, and microservices. Ships systems that process hundreds of millions of dollars in transactions, including ETL automation that eliminates 100+ hours of manual work per quarter while holding error rates under 1%. Increasingly focused on AI agent tooling: least-privilege automation with human-in-the-loop approval on anything outward-facing. Thrives in fast-moving, high-ownership environments.

Experience

Software Engineer

December 2022 – Present

Paycom · Irving, TX

- Architected and implemented an internationalization framework for an enterprise-scale tax API using C# and .NET, enabling automated multi-country payroll controls processing across 5 countries with high compliance accuracy.
- Led end-to-end design and delivery of mission-critical regulatory compliance controls for domestic garnishment, tax withholding, and earnings regulations within the core payroll engine, processing hundreds of millions of dollars in annual transactions while maintaining 100% audit compliance.
- Independently architected and deployed a full-SDLC automated ETL pipeline to parse, validate, and transform 50+ state-specific SUTA tax rate files into controlled system updates, eliminating 100+ hours of manual processing per quarter and reducing error rates to under 1%.
- Proactively monitor production systems, triage incidents, and implement reliability improvements, reducing downtime and increasing application availability for enterprise-scale payroll infrastructure.
- Collaborate cross-functionally with product, QA, and engineering teams to gather requirements, design system architecture, and conduct code reviews that uphold reliability and security standards.

Enterprise Architect Developer (Internship)

May 2021 – December 2021

Valero Energy Corporation · San Antonio, TX

- Developed a mission-critical SAP GUI application in ABAP for the Accounts Payable team, supporting processing for 90% of the company's U.S. financial transactions with high reliability and accuracy.
- Engineered enterprise-grade software within a Fortune 500 environment, translating complex operational requirements into scalable, production-ready systems.
- Documented software components, integration flows, and technical specifications for cross-team knowledge transfer.

Projects

Kowalski `Personal AI Agent System`

- Stood up a “Second Brain” long-term memory system in Obsidian, semantically organized across people, projects, security, and finance, so the agent retains decision history and context across sessions instead of starting cold each time. Paired with a 15+ tool integration layer (Gmail, Calendar, Drive, GitHub, voice synthesis) for end-to-end task handling.
- Extended the agent into a financial decision-support tool that tracks portfolio positions, maintains investment-thesis notes, and summarizes account activity from email, while keeping trade execution and approval strictly manual to preserve financial-control discipline.

AI Gaming Assistant `Full-Stack Desktop Application`

- Built a cross-platform desktop gaming companion (Tauri 2 + Rust, React/TypeScript; FastAPI backend on Supabase + Upstash Redis) with streaming AI chat, automatic session tracking across 80+ games, Discord integration, and a content feed.
- Engineered a dual-mode AI inference router with failover across cloud providers (Groq, Together AI) and local models (Ollama); shipped Stripe subscription billing and a game-aware notification engine, backed by 238 automated tests across Python, TypeScript, and Rust.

Technical skills

LANGUAGES	Python, C, C++, C#, JavaScript, SQL, Rust, PHP, ABAP
BACKEND & APIS	.NET, Node.js, FastAPI, RESTful APIs
FRONTEND	React, TypeScript SPAs, Tailwind CSS
DATABASES	PostgreSQL, SQLite, MySQL, query optimization, data pipeline development
DEVOPS & CLOUD	Git, Docker, AWS EC2, Linux/UNIX
PRACTICES	Agile/Scrum, TDD, code reviews, system design, production monitoring

Education

B.S. Computer Science `Graduated December 2022`

`The University of Texas at San Antonio`

Concentrations in Software Engineering & Data Science.

Additional

- ACM ICPC member; strengthened algorithmic problem-solving through team-based competitive programming.
- Experienced public speaker with strong written and verbal communication skills.
- Bias toward full ownership and rapid iteration in high-intensity, mission-driven environments.